

Inference as Currency

A monetary theory of the AI cycle, in three pages.

May 2026 · A working paper

Microsoft cancelled its internal Claude Code seats this May. Microsoft owns Azure, holds 19% of OpenAI, and sells more enterprise software than anyone — the most-aligned buyer for token-priced AI that exists. If the math doesn't clear there, the buyer-side problem is structural. Same month: Uber burned its entire 2026 AI budget in four months; GitHub deprecated flat-rate Copilot; Anthropic moved from a token meter to opaque weekly caps; OpenAI offered \$2M in tokens for equity to every founder at a YC dinner.

The consensus view writes much larger numbers. Trillion-dollar lab revenues by 2030, with “AI finance ministers and central bankers” driving 10% GDP growth in the developing world;^[1] labs priced on Netflix-and-Spotify comparables, durable margins, personalisation moat, the commoditisation thesis “more or less irrelevant”;^[2] an \$800B market the lab's seed investors call “a gross underestimate.”^[3] None of those projections survives this month's data.

AI labs are private currency issuers. The currency is the token. The underlying asset is the *verifier substrate* — the human or software that decides whether AI work was done correctly. Token value behaves like nineteenth-century private banknote value: face inside the issuer's perimeter, discounted with distance, vulnerable to runs when redemption fails. The rent on a token tracks the K-complexity of verification in the buyer's domain, not model capability and not token volume.

The token

A token is a claim on inference, redeemable through a substrate: a compiler accepts the code, a partner signs the brief, a clinician signs the chart, a regulator approves the filing.

Kolmogorov complexity of verification, $K(\text{verify})$, is the length of the shortest program that maps an output to a yes-or-no judgment given the verifier's information set I_v .^[4]

$$K(\text{verify} \mid t, I_v) = \min_p |p| \quad \text{s.t.} \quad U(p, y, t, I_v) = \mathbb{1}[y \models t].$$

For a compiler, I_v is language semantics plus a test harness — $K(\text{verify})$ is small. For a partner reviewing a brief, I_v is the assigned judge's record, opposing counsel's playbook, the client's risk tolerance, the firm's strategy — $K(\text{verify})$ is enormous and mostly tacit. The rent on a token tracks $K(\text{verify})$, not capability:

$K(\text{verify})$	Verifier	Rent accrues to
Small	Compiler, test harness, schema	Compute owner
Medium	Domain software, structured audit	Substrate owner
Large	Credentialed human, signed + indemnified	The signature

Discount by distance

A token from Anthropic redeems at face inside Claude Code. The same token used in a third-party app without Anthropic's vertical integration redeems at a discount; in a generic domain without a productisable verifier it doesn't redeem at all.

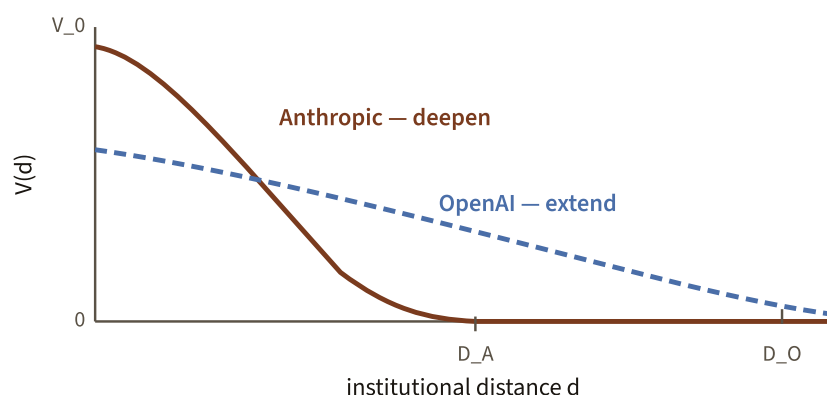
The pattern is nineteenth-century banknote economics.^[5] A note from an Indiana bank traded at face in its home county, at a discount in the next state, and as wallpaper across the Mississippi. Free Banking America (1837–1863) produced 800+ private currency issuers; the National Banking Acts of 1863–1864 unified the system. Formally:

$$V(d) = V_0 \cdot e^{-\lambda d}.$$

d is institutional distance from the lab's perimeter and λ is the lab-specific discount rate. The lab's strategic problem is

$$\min_{\lambda(\cdot), D} \int_0^D \lambda(d) \, dd \quad \text{s.t.} \quad C(\lambda, D) \leq B,$$

a perimeter-investment budget B on cost surface $C(\cdot)$.^[6] Two corner solutions: deepen (low λ , narrow D) or extend (moderate λ , wide D).



Redemption value vs. institutional distance. Anthropic minimises λ inside a narrow perimeter; OpenAI accepts a higher λ in exchange for wider D .

Two strategies

Anthropic deepens. Direct enterprise contracts in regulated verticals — clinical decision support, claims adjudication, regulatory affairs — account for roughly 80% of revenue. First-party harnesses (Claude Code, Coefficient) capture token volume inside the lab. OAuth restrictions pull telemetry inside the perimeter. The bet: $K(\text{verify})$ stays high; residual rent accrues to substrate owners.

OpenAI extends.^[7] Multi-vendor supply with Microsoft, Amazon, Google flattens hyperscaler dependence. A horizontal API business prioritises developer adoption. The \$2M YC token offer buys equity in tomorrow's app businesses paid in OpenAI's own currency. The bet: $K(\text{verify})$ falls; volume across many small substrates dominates depth in any one.

	Anthropic — deepen	OpenAI — extend
Perimeter	narrow, high-investment	wide, low-investment
Revenue mix	~80% enterprise verticals	API + capacity reservations
$K(\text{verify})$ bet	stays high	falls
Token sink	first-party harnesses	third-party apps + YC equity
Failure mode	vertical gets disrupted	volume base commoditises

Both are autarky. The labs are not competing for the same customer. They are betting on opposite futures of the same parameter.

Bank runs

Microsoft's cancellation is a single-buyer redemption failure — a depositor turned away. The bank survives the withdrawal. Information about whether it could honour at face propagates. Other large enterprises run the same cost-benefit math. Enough simultaneous cancellations and the lab faces a coordinated run. Let R_b be the cost to construct buyer b 's redemption infrastructure and V_b the realised per-token value; b cancels when

$$R_b > V_b = V_0 \cdot e^{-\lambda d_b}.$$

The threshold is endogenous in λ . Weekly rate caps and priority-service contracts push pricing into bilateral capacity reservations — the retreat Free Banking-era banks made when they ran out of specie.^[8]

Episode	Backing	Outcome
TerraUSD (May 2022)	algorithmic — none	\$45B destroyed, 1 week
USDC (Mar 2023)	full but SVB-exposed	depeg, 2 days, restored
Tether (2016–18)	27.6% (NY AG settlement)	survived; never fully audited

Private currency without sufficient backing fails not gradually but suddenly, when buyer confidence cracks.

Resolution

Path	Mechanism	Analog	Trigger
State imposition	Regulated-issuer framework; USG consortium plausible ^[9]	National Banking Acts 1863–64	Visible run + geopolitics

Path	Mechanism	Analog	Trigger
Serial failure	Smaller labs fail or merge; survivors compress to defense-contractor multiples	UK 19th-century banking	Time-gap stress
Multipolar fragmentation	US, Chinese, EU, Indian, Gulf systems; cross-border verifier-credentialing the binding institution	Pre-1914 gold-standard era	Status quo extends

Probability is roughly equal across the three on a five-year horizon. The realised outcome will be a combination.

What this means

Lab equity is priced on platform-software comparables. Realised revenue and margin look like cost-pass-through infrastructure. Multiple compression from platform to monetary-institution multiples is the modal outcome, not a tail risk.

The institutions that benefit are verifier-substrate owners — credentialing bodies, audit firms, vertical software that owns the substrate in its domain (Halter, Waymo, OpenEvidence, Sardine, Abridge). Hardware and energy infrastructure benefits regardless of resolution path.

The credentialed human at the bottom of the signature chain is the closest thing the cycle has produced to specie reserves. She holds value across every redemption scenario; no lab can capture her without integrating into the substrate where her signature lives. The wage of the doctor, the lawyer, the engineer, the underwriter, the pilot rises in real terms over five years. The middle of the labour market — the analyst, the coordinator, the entry-level associate — compresses out of every column simultaneously. The political debate about AI and labour has been sighting the wrong tier.

The cycle has been pricing the wrong substrate.

1. Dario Amodei, “Machines of Loving Grace” (Oct 2024). Targets 20% annual GDP growth in the developing world, with AI replicating East Asian-style ~10% growth via “AI finance ministers and central bankers.” Trillion-dollar revenues by 2030 are presented as plausible under the bull-case capability trajectory. ↩
2. Dan Sundheim, D1 Capital, Invest Like the Best Ep. 460 (Feb 2026): LLM businesses combine fixed-asset economics (Netflix content) with personalisation-driven stickiness (Spotify playlists); gross margins are durable; the commoditisation thesis “is more or less irrelevant.” The strongest published counter to the framing here. ↩
3. Anjney Midha (Anthropic seed investor, ex-a16z): “in the early days, a couple Anthropic cofounders and I started doing weekly working sessions for our first fundraiser we needed a coherent view of our market size and arrived at an answer that felt reasonable: ~\$800B/year as Andrej has discovered, that was a gross underestimate” — X.com, Oct 2025. Dwarkesh Patel adds that founders and early investors of frontier labs will “own a galaxy.” ↩
4. The K(verify) construction extends conditional Kolmogorov complexity (Li & Vitányi, *An Introduction to Kolmogorov Complexity*, 4th ed., Springer 2019) using a verifier’s tacit information set as the conditioning variable. Underlying economic machinery: Akerlof, “Lemons” (*QJE* 1970); Holmström, “Moral Hazard and Observability” (*Bell J. Econ.* 1979); Bergemann & Bonatti, “Markets for Information” (*Annu. Rev. Econ.* 2019). ↩
5. Hugh Rockoff, “The Free Banking Era: A Reexamination,” *Journal of Money, Credit and Banking* 6:2 (1974), pp. 141–167. For the political economy of the resolution, Calomiris & Haber, *Fragile by Design* (Princeton, 2014). The discount-by-distance pattern was geometric in observable broker quotes by 1850. ↩
6. Ronald Coase, “Durability and Monopoly,” *J. Law & Econ.* 15 (1972). Board & Pycia, “Outside Options and the Failure of the Coase Conjecture,” *American Economic Review* 104:2 (Feb 2014). The labs’ eighty-percent gross margins are the Board-Pycia selection signature, not a moat. ↩
7. Joel Spolsky, “Strategy Letter V: Commoditize Your Complement,” joelonsoftware.com (June 2002). Each layer’s complement is what it tries to cheapen. Hamilton Helmer, *Seven Powers* (2016), on counter-positioning: the labs’ token meter is counter-positioned against the SaaS flat-rate model, and SaaS cannot retrofit per-token pricing without breaking customer relationships. ↩
8. TerraUSD / LUNA collapse, May 2022 — algorithmic peg failed; ~\$45B value destroyed in a week. USDC depeg, March 2023 — Silicon Valley Bank held Circle’s reserves. Tether reserve fraction during 2016–18 was 27.6% per the NY Attorney General’s 2021 settlement. Three independent demonstrations that private currencies without sufficient backing fail discontinuously when confidence cracks. ↩
9. Leopold Aschenbrenner, “Situational Awareness: The Decade Ahead” (June 2024), section “The Project”. Predicts USG absorption of frontier labs into a Manhattan-Project-style consortium by 2027–28 under Lockheed/Boeing-type defense contracting arrangements, triggered by capability + geopolitical pressure. ↩